

Nabil Rakaiza Abror | nabilraka1234@gmail.com | +65 8917 6960 |
github.com/nabilrakaiza | linkedin.com/in/nabilrakaiza/ | nabilrakaiza.vercel.app

Education

National University of Singapore Aug 2024 – Jun 2028
B.Sc. (Hons.) in Data Science and Analytics with Second Major in Computer Science
GPA: 4.69/5.00 (Highest Distinction)
Honors: Top Student — CS1010S (Programming Methodology), IT1244 (AI: Technology & Impact)

Work Experience

Data Science Intern May 2025 – July 2025
Quantum Teknologi Nusantara Indonesia

- Built and deployed 3 AI systems using Python and LLM frameworks, including a CVRPTW route optimizer, an AI graphic designer, and an ontology-based news relationship engine
- Improved LLM output relevance and consistency by ~20% through prompt decomposition, structured outputs, and evaluation loops
- Collaborated with engineers and product stakeholders to translate business needs into data-driven solutions

Undergraduate Teaching Assistant (CS2030) Jan 2026 – Present
National University of Singapore Singapore

- Mentored 20+ students weekly in object-oriented design, immutability, and functional abstractions in Java
- Debugged and reviewed student codebases, identifying design flaws, improper abstractions, and state-related bugs
- Guided students through complex project milestones, emphasizing modularity, testability, and clean architecture

Projects

Automatic Survey Analyzer | Machine Learning Project / [GitHub Link](#)

- Built an end-to-end automated survey analysis pipeline in Python for insight extraction and survey optimization
- Trained an XGBoost model to identify key factors behind incomplete survey responses, using SHAP values for interpretability
- Applied DBSCAN clustering and LLM-based column identification to automate demographic segmentation and respondent profiling
- Generated actionable insights to guide survey design improvements

Oil Temperature Transformers Time Series Prediction | Machine Learning Project / [GitHub Link](#)

- Developed and evaluated 12+ machine learning and deep learning models for time-series temperature prediction
- Performed data cleaning, feature engineering, anomaly detection, and model comparison to improve robustness

GrindHub | Mobile Application Project / [GitHub Link](#)

- Built a full-stack productivity app and integrated an LLM-based chatbot for study assistance
- Designed clean interfaces between backend services and AI components using Flask and LangChain
- Explored applied LLM usage in a real user-facing product

Organizational Experience

Perhimpunan Indonesia NUS (PINUS) | Senior Developer Sep 2025 - Present

- Led a team of 2 junior developers in building the PINUS Forms website, utilizing Next.js and Tailwind CSS to deliver a responsive and efficient user experience.

Skills

Programming: Python, R, SQL/PostgreSQL, Java, JavaScript, TypeScript, C++
ML & Data: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, LangChain, LangGraph, Gemini, OpenAI
Frameworks: React Native, React, Node.js, Express, Next.js, Flask
Other Tools: Figma, GitHub, Power BI, Tableau, Microsoft Excel, Microsoft Azure, Supabase
Languages: English (Fluent), Indonesian (Native)